

INTERNSHIP REPORT

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FOSSEE eSim SEMESTER LONG INTERNSHIP – AUTUMN 2026

TASK 5 – AUTOMATED TOOL MANAGER

eSimMate

Automated Tool Installation, Version Management,
Compatibility Checking and System Diagnostics for eSim

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Submitted By

DHARANIDHARAN T.

B.Tech. COMPUTER SCIENCE AND BUSINESS SYSTEMS

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Sri Eshwar College of Engineering
(An Autonomous Institution)

Kinathukadavu (Tk), Coimbatore - 641 202, Tamil Nadu

Approved by AICTE, New Delhi and Affiliated to Anna University, Chennai



DECLARATION

I hereby declare that the technical report entitled "**ESIMMATE: AUTOMATED TOOL INSTALLATION, VERSION MANAGEMENT, COMPATIBILITY CHECKING AND SYSTEM DIAGNOSTICS FOR ESIM**" submitted for Task 5 (Automated Tool Manager) of the FOSSEE eSim Semester Long Internship – Autumn 2026 is an authentic record of original engineering work carried out by me. This work represents a student project developed for the screening evaluation of the FOSSEE eSim Semester Long Internship – Autumn 2026. All software components, algorithms, test suites, and empirical results documented in this report represent actual implemented and verified artifacts residing in the project repository <https://github.com/dharanidh-02/eSimMate>. Appropriate academic and technical citations have been given where external libraries, tools, or specifications have been referenced.

DATE: 20/08/2026

[DHARANIDHARAN T]

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I also acknowledge the global open-source Electronic Design Automation (EDA) community—including the development teams of KiCad, Ngspice, GHDL, OpenModelica Compiler (OMC), Verilator, and FreeCAD—whose foundational software tools enable open-source circuit design and simulation.

ABSTRACT

eSim, an open-source Electronic Design Automation (EDA) suite developed by FOSSEE at IIT Bombay, depends on an ecosystem of external open-source simulation and design engines including KiCad, Ngspice, GHDL, OpenModelica Compiler (omc), and Verilator. Installing, verifying, and maintaining these heterogeneous tools across different operating systems presents critical challenges: platform package manager fragmentation, version compatibility drift, missing binary paths, and security risks from unverified privileged execution scripts. To address these problems, this report presents eSimMate, a modular, configuration-driven, cross-platform automated tool and dependency management framework developed in Python 3.10–3.12. eSimMate provides an extensible YAML tool registry, non-intrusive binary and PATH detection, semantic version extraction using safe subprocess execution with regex filtering, multi-bound compatibility evaluation ([min_version .. max_version]), and polymorphic package manager adapters (apt, winget, choco). eSimMate introduces a resilient official archive fallback strategy (MANUAL_DOWNLOAD) for tools unavailable in upstream package repositories (such as Windows Ngspice), combining automated archive download, HTTP validation, multi-format extraction (.7z/.zip), executable discovery, and local path registration. eSimMate enforces strict execution safety invariants (shell=False, zero os.system()), interactive user elevation confirmation, dry-run previews, and persistent rotating audit logs. The system achieves 87% overall statement code coverage across 1,189 statements with 93 automated unit and integration tests passing in 8.10–14.02 seconds. Real-world validation on a Windows 10/11 environment (build 26200, x86_64, Python 3.11.9, WinGet) verified automated discovery and compatibility validation of Ngspice version 46 (ngspice_con.exe) in 0.062 seconds.

EXECUTIVE SUMMARY

Problem: Modern open-source EDA platforms rely on complex multi-tool execution pipelines. eSim integrates KiCad for schematic capture and PCB layout, Ngspice for analog circuit simulation, GHDL for digital VHDL simulation, OpenModelica for mathematical modeling, and Verilator for Verilog co-simulation. Because these external tools evolve on independent release cycles, users frequently encounter broken simulations due to version mismatches, missing binary paths, or package manager repository failures.

Motivation: Existing installation approaches rely on monolithic platform scripts (e.g. bash scripts on Linux or bundled installers on Windows) that execute unverified commands, lack pre-execution dependency inspection, fail when upstream packages are dropped, and risk system misconfiguration.

Proposed Solution: eSimMate replaces brittle scripts with a modular, declarative Python architecture. Tool specifications reside in configs/tools.yaml. System detection, version extraction, package manager abstraction, installation execution, logging, and CLI presentation are decoupled into independent subsystems.

Key Contributions:

- **Dynamic, Configuration-Driven Tool Registry:** Zero hardcoded Python tool classes; tools, binary aliases, and version bounds are loaded dynamically from YAML files.
- **Polymorphic Package Manager Abstraction:** Unified adapter interface for apt, winget, choco, and custom scripts.
- **Official Archive Fallback Strategy:** Solves missing upstream repository packages (such as Windows Ngspice) through end-to-end download, HTTP validation, 7-Zip extraction, executable discovery, and managed path registration.
- **Safety & Zero-Surprise Privilege Handling:** Eliminates `os.system()` and `shell=True`, enforcing parameterized argument lists, dry-run previews, and user elevation prompts.
- **5-Stage System Diagnostics (esimmate doctor):** Comprehensive environment, package manager, tool health, and environment variable inspection emitting actionable remediation guidance.
- **Empirical Validation & Test Coverage:** 93 automated unit and integration tests passing in 8.10–14.02 seconds with 87% statement code coverage (1,189 statements).

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LIST OF ABBREVIATIONS

API — API: Application Programming Interface

CLI — CLI: Command Line Interface

CPU — CPU: Central Processing Unit

EDA — EDA: Electronic Design Automation

FOSSEE — FOSSEE: Free/Libre and Open Source Software for Education

GHDL — GHDL: VHDL Simulator (Gnu Hardware Description Language)

GUI — GUI: Graphical User Interface

HDL — HDL: Hardware Description Language

HTTP — HTTP: Hypertext Transfer Protocol

IIT — IIT: Indian Institute of Technology

LOC — LOC: Lines of Code

NMEICT — NMEICT: National Mission on Education through ICT

OMC — OMC: OpenModelica Compiler

OS — OS: Operating System

PCB — PCB: Printed Circuit Board

PDK — PDK: Process Design Kit

PEP — PEP: Python Enhancement Proposal

RAM — RAM: Random Access Memory

SPICE — SPICE: Simulation Program with Integrated Circuit Emphasis

UAC — UAC: User Account Control

VHDL — VHDL: VHSIC Hardware Description Language

WinGet — WinGet: Windows Package Manager

YAML — YAML: YAML Ain't Markup Language

INTRODUCTION

1.1 Background

Electronic Design Automation (EDA) comprises software algorithms, modeling engines, and graphical workflows necessary for designing integrated circuits and Printed Circuit Boards (PCBs). Because complete circuit engineering requires analog SPICE simulation, digital hardware description compilation, multi-domain system modeling, and mechanical 3D modeling, modern open-source EDA platforms orchestrate independent, specialized computational backends.

1.2 eSim Overview

eSim (formerly Oscad) is an open-source EDA tool developed by the FOSSEE (Free/Libre and Open Source Software for Education) project at the Indian Institute of Technology Bombay (IIT Bombay), funded by the National Mission on Education through ICT (NMEICT), Ministry of Education, Government of India. eSim provides an integrated environment for circuit design, schematic capture, SPICE simulation, mixed-mode simulation, and PCB layout.

1.3 FOSSEE Context

FOSSEE promotes the adoption of open-source software tools in educational institutions across India. As part of this mission, eSim integrates several standalone open-source software packages to offer an alternative to proprietary commercial EDA software.

1.4 External Tool Dependencies

eSim depends on the following external software engines:

- KiCad EDA: Schematic capture, symbol library management, footprint assignment, and PCB layout design.
- Ngspice: Mixed-level, mixed-signal SPICE circuit simulation engine for DC, AC, and transient analysis.
- GHDL: VHDL analyzer and simulator enabling digital hardware description verification and mixed-signal co-simulation (NGHDL).
- OpenModelica Compiler (omc): Multi-domain mathematical equation modeling and system simulation.
- Verilator: Fast Verilog/SystemVerilog simulator compiling HDL models into native C++ executables.
- FreeCAD: 3D CAD parametric modeler for mechanical enclosure validation and 3D PCB visualization.

1.5 Motivation

Because each external tool evolves on an independent release schedule with distinct binary naming conventions, search paths, and installer mechanisms across operating systems (Linux vs. Windows), users frequently experience installation failures, missing executables on \$PATH, and version incompatibilities that disrupt circuit simulation workflows.

1.6 Need for Automated Tool Management

Manual dependency management is error-prone. Users require an automated utility that inspects the host platform, validates tool presence and version compatibility, resolves OS package managers, and executes installations safely with elevation prompts and dry-run previews.

1.7 Project Scope

The scope of eSimMate encompasses automated system platform detection, dynamic external tool discovery, semantic version extraction, multi-bound compatibility evaluation ([min_version .. max_version]), package manager abstraction (apt, winget, choco), resilient archive fallback installation, dry-run previews, action audit logging, and a 5-stage health diagnostic CLI.

1.8 Contributions

This project contributes: (1) an extensible YAML tool registry; (2) safe version extraction supporting interactive console banners; (3) polymorphic package manager adapters; (4) an automated official archive fallback strategy; (5) strict process execution safety without shell=True; and (6) a test suite of 93 tests achieving 87% code coverage.

PROBLEM STATEMENT AND REQUIREMENTS

2.1 Problem Statement

To design and implement a Python-based Automated Tool & Dependency Manager (eSimMate) for eSim that accurately discovers installed EDA binaries, verifies semantic version bounds, abstracts cross-platform package managers, executes safe non-destructive installations with user confirmation, handles missing package-manager repositories via official archive fallbacks, and delivers an expressive terminal user interface with comprehensive audit logging.

2.2 Problem Analysis

EDA workflows break when underlying tools are missing or incompatible. For example, eSim circuit netlists generated by KiCad must be formatted to match the SPICE syntax expected by the installed Ngspice release. If Ngspice is missing from \$PATH or an outdated build is present, simulation fails during runtime without clear diagnostics.

2.3 Challenges

1. Binary Naming Variations: Executables differ across operating systems (e.g. ngspice on Linux vs. ngspice_con.exe / ngspice.exe on Windows).
2. Interactive Console Hanging: Querying versions on tools like Ngspice may open an interactive prompt rather than exiting, causing standard subprocess calls to hang.
3. Package Repository Gaps: Windows WinGet repositories frequently lack specialized scientific tools like Ngspice.
4. Execution Safety: Monolithic scripts invoke unverified root/administrator commands blindly.

2.4 Objectives

- Automated Tool Detection across \$PATH and standard search paths.
- Semantic Version Verification against configurable bounds.
- Polymorphic Package Manager Abstraction (apt, winget, choco).
- Resilient Archive Fallback Strategy for missing repository packages.
- Strict Security and Privilege Safety (zero shell=True, dry-run previews).
- Comprehensive 5-Stage Diagnostics and Persistent Action Logging.

2.5 Functional Requirements

TABLE I
ESIMMATE FUNCTIONAL REQUIREMENTS SPECIFICATION

ID	Category	Functional Requirement Description	Module
FR-1.1	Detection	Scan system \$PATH and candidate directories for registered tool binaries.	detector.py
FR-1.2	Detection	Detect host OS, distribution, CPU architecture, bitness, and Python runtime.	detector.py
FR-2.1	Versioning	Safely query tool versions via subprocess execution with timeouts.	version_manager.py
FR-2.2	Versioning	Extract semantic version strings using regex matching from stdout/stderr.	version_manager.py
FR-2.3	Versioning	Classify tool status: COMPATIBLE, NOT_INSTALLED, OUTDATED, NEWER_VERSION, etc.	compatibility.py
FR-3.1	Installation	Resolve active package manager adapter (apt, winget, choco) or fallback.	installer.py
FR-3.2	Installation	Support Dry-Run Mode (--dry-run) displaying exact commands and target paths.	installer.py
FR-3.3	Installation	Require explicit confirmation before executing privileged operations.	installer.py
FR-3.4	Installation	Execute official archive fallback download, extraction, and path registration.	installer.py
FR-4.1	UI / CLI	Provide CLI commands: list, check, install, update, doctor, logs, config.	cli.py
FR-4.2	UI / CLI	Support --json flag on query commands for automated scripting.	cli.py
FR-5.1	Logging	Maintain persistent rotating audit logs in logs/esimmate.log.	logger.py

2.6 Non-Functional Requirements

TABLE II
ESIMMATE NON-FUNCTIONAL QUALITY ATTRIBUTES

ID	Quality Attribute	Technical Invariant & Specification
NFR-1	Safety & Security	Prohibit os.system() and shell=True. Commands must pass argument lists to subprocess.run().
NFR-2	Zero Silent Elevation	Never execute elevated commands (sudo, Administrator) without user confirmation.
NFR-3	Performance	Tool detection and compatibility checks must complete in <3.0 seconds for the entire suite.
NFR-4	Subprocess Safety	Process execution must enforce a 5-second timeout on version queries and clean up spawned processes.
NFR-5	Zero Hardcoded Paths	No user-specific or machine-specific absolute paths hardcoded in source code.
NFR-6	Test Isolation	Unit test suite must run with 100% mocked package managers and subprocesses.

2.7 FOSSEE Task 5 Requirements

TABLE III
FOSSEE TASK 5 SCREENING REQUIREMENTS BASELINE

Requirement ID	Requirement Title	Scope Priority	eSimMate Implementation Strategy
Requirement 1	Tool Installation Management	Primary	Full installation pipeline (apt, winget, choco, manual_download fallback), dynamic availability check, dry-run mode, confirmation safety.
Requirement 2	Update and Upgrade System	Secondary	VersionManager detects outdated tools; update command identifies tools requiring action and invokes package manager workflows. (Independent index sync is partial).
Requirement 3	Configuration Handling	Secondary	PyYAML configuration management (configs/config.yaml, configs/tools.yaml), user overrides in ~/.esimmate/tools.d/.
Requirement 4	Dependency Checker	Secondary	esimmate check and esimmate doctor evaluate tool presence, versions, and environment variables against compatibility bounds.
Requirement 5	User Interface	Primary	Typer/Rich CLI interface featuring colored status tables, interactive prompts, log viewers, and machine-readable --json output.
Requirement 6	Additional Features & Safety	Secondary	Rotating file audit logs (logs/esimmate.log), parameter safety (shell=False), 93 unit/integration tests, 87% coverage.

2.8 Requirement Traceability

Every functional and non-functional requirement maps directly to dedicated Python source modules and corresponding unit tests in the tests/ directory.

EXISTING SYSTEM AND MOTIVATION

3.1 Existing Manual Approach

The existing eSim setup process relies on monolithic installation scripts on Linux (install-eSim.sh) and bundled executable installers on Windows.

3.2 Manual Installation Workflow

Users execute monolithic installation scripts that sequentially attempt to install packages via system package managers or unpack pre-bundled binary archives into hardcoded directories (e.g. C:\FOSSEE\eSim\).

3.3 Problems with Manual Management

- **No Pre-Execution Inspection:** Does not verify whether tools already exist on \$PATH or whether installed versions meet compatibility bounds.
- **Fragile Repository Dependencies:** Script execution aborts when package repositories fail or drop package names.
- **Privilege Risks:** Scripts require administrative/root execution unconditionally, risking modification of unrelated system packages.

3.4 Limitations

Existing scripts lack dry-run capabilities, structured JSON output for CI/CD automation, dynamic multi-candidate binary search, and comprehensive diagnostic health reports.

3.5 Motivation for eSimMate

eSimMate was created to replace brittle monolithic scripts with a modular, declarative, and secure Python framework that automates detection, version validation, and installation.

PROPOSED SYSTEM

4.1 Overview

eSimMate provides a modular Python 3.10–3.12 application designed according to SOLID principles. The architecture cleanly separates system detection, tool registry metadata, version parsing, package manager abstraction, installation execution, logging, and CLI presentation.

4.2 eSimMate Architecture

The system architecture comprises 10 decoupled subsystems interacting through well-defined APIs and typed dataclasses.

4.3 Design Principles

- Single Responsibility Principle (SRP): Each module encapsulates exactly one responsibility.
- Open-Closed Principle (OCP): New tools and package managers can be added without modifying existing Python codebase.
- Liskov Substitution Principle (LSP): All package managers implement `AbstractPackageManager` seamlessly.
- Interface Segregation Principle (ISP): Presentation layers consume only relevant high-level manager methods.
- Dependency Inversion Principle (DIP): Managers depend on abstractions rather than concrete implementations.

4.4 Configuration-Driven Architecture

Tool definitions, binary aliases, regex extraction rules, package IDs, and compatibility ranges reside entirely in `configs/tools.yaml`.

4.5 Modular Architecture

Subsystems (detector, registry, version_manager, installer, package_manager, logger, cli) are decoupled and independently testable.

4.6 Safety-First Installation

Prohibits `os.system()` and `shell=True`, enforcing parameterized argument lists, dry-run previews, and user elevation prompts.

4.7 Fallback Installation Strategy

Implements automated archive download, integrity verification, extraction, and managed path registration when package managers lack upstream packages.

SYSTEM ARCHITECTURE

5.1 High-Level Architecture

The high-level architecture of eSimMate is depicted in Figure 1.

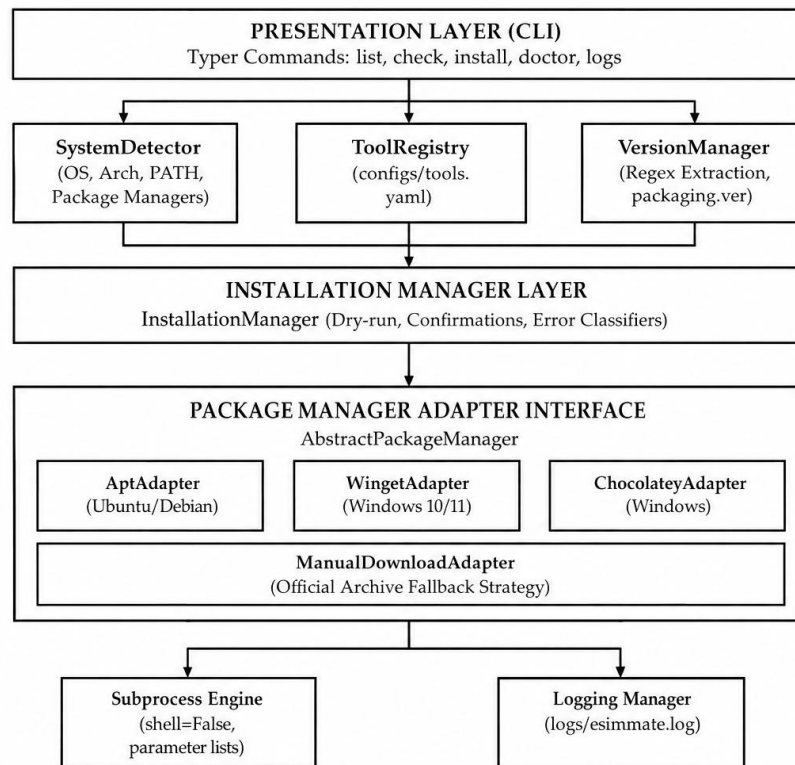


Figure 1. Overall eSimMate Architecture

5.2 CLI Layer

Built using Typer and Rich, the CLI layer provides structured subcommands (list, check, install, update, doctor, logs, config).

5.3 Tool Registry

Dynamically loads YAML definitions and populates ConfigurableTool instances.

5.4 Detector

The SystemDetector inspects OS kernel, release, CPU architecture (x86_64, arm64), bitness, and available package managers.

5.5 Version Manager

Executes version query commands with timeouts, parses stdout/stderr with regex patterns, and supports interactive banner scraping.

5.6 Compatibility Engine

Evaluates installed version tuples against min_version and max_version bounds.

5.7 Installer

Coordinates package manager selection, elevation checks, dry-run previews, subprocess execution, and official archive fallbacks.

5.8 Package Manager

Base adapter interface implemented by AptAdapter, WingetAdapter, ChocolateyAdapter, and ManualDownloadAdapter.

5.9 Configuration Manager

Loads user preferences and tool definitions from YAML files.

5.10 Logger

Maintains rotating persistent audit logs in logs/esimmate.log.

5.11 Data Flow

Figure 2 illustrates the complete end-to-end data flow across subsystems.

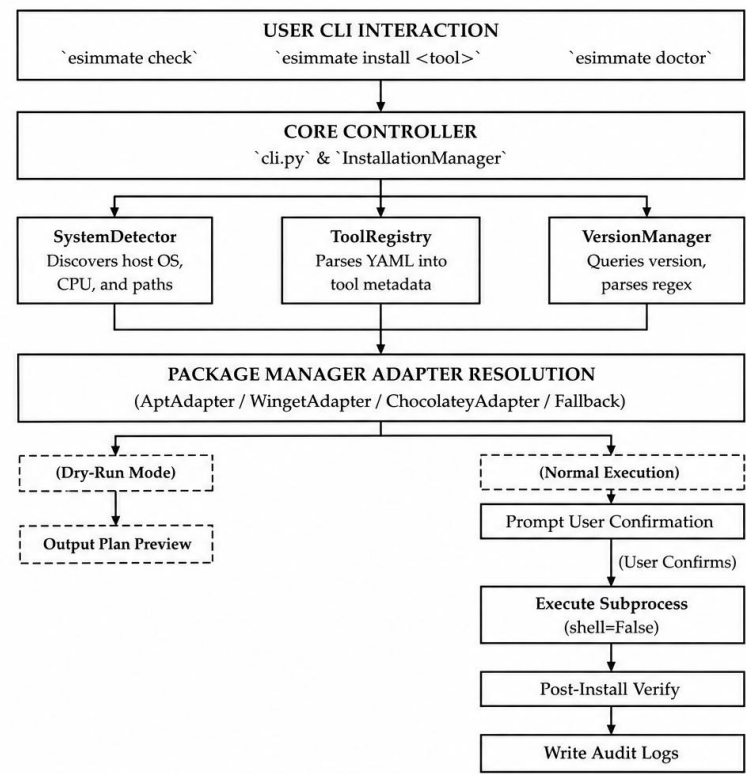


Figure 2. Complete End-to-End Data Flow

5.12 Component Interaction

Each component communicates via strongly typed dataclasses (`SystemInfo`, `VersionCheckResult`, `InstallationResult`), ensuring loose coupling.

TOOL REGISTRY AND CONFIGURATION

6.1 Registry Design

The ToolRegistry dynamically instantiates ConfigurableTool objects by parsing YAML files. It supports directory scanning for custom user tool definitions at `~/.esimmate/tools.d/`.

6.2 tools.yaml

The primary database `configs/tools.yaml` defines metadata, executables, regex patterns, version bounds, and package manager identifiers for all eSim external tools.

TABLE IV
REGISTERED EXTERNAL TOOLS

Tool ID	Tool Name	Mandatory	Category	Executables (Linux/Win)	Target Range	Package Identifiers
kicad	KiCad EDA	Yes	schematic_pcb	kicad / kicad.exe	[6.0.0 .. 8.0.99]	apt: kicad winget: KiCad.KiCad choco: kicad
ngspice	Ngspice Simulator	Yes	simulation	ngspice / ngspice_con.exe	[34 .. 46]	apt: ngspice winget: Ngspice.Ngspice Fallback: SourceForge 7z
ghdl	GHDL VHDL Simulator	No	hdl_vhdl	ghdl / ghdl.exe	[2.0.0 .. 4.1.0]	apt: ghdl choco: ghdl
openmodelica	OpenModelica (omc)	No	modeling	omc / omc.exe	[1.19.0 .. 1.22.99]	apt: openmodelica winget: OpenModelica.OpenModelica
verilator	Verilator	No	hdl_verilog	verilator / verilator.exe	[4.200 .. 5.999]	apt: verilator
freecad	FreeCAD Modeler	No	3d_cad	freecad / FreeCAD.exe	[0.19.0 .. 1.0.0]	apt: freecad winget: FreeCAD.FreeCAD choco: freecad
python	Python 3 Runtime	Yes	runtime	python3 / python.exe	[3.10.0 .. 3.12.99]	apt: python3 winget: Python.Python.3.11 choco: python

6.3 config.yaml

Global user preferences, log verbosity, and custom search paths reside in `configs/config.yaml`.

6.4 Tool Metadata

Encapsulated by ToolMetadata dataclass: `id`, `name`, `category`, `mandatory`, `purpose`, `executables`, `compatibility`, `version_check`, `supported_platforms`, `package_managers`, `default_search_paths`, `environment_variables`, `optional_config`, and `executable_candidates`.

6.5 Version Constraints

TABLE V
TOOL VERSION REQUIREMENTS

Tool ID	Minimum Version	Recommended Version	Maximum Version	Compatibility Rationale
kicad	6.0.0	7.0.10	8.0.99	eSim 2.5 schematic netlist generation requires KiCad 6–8.
ngspice	34	38	46	SPICE simulation engine; verified up to release 46.
ghdl	2.0.0	3.0.0	4.1.0	Digital VHDL analysis and co-simulation.
openmodelica	1.19.0	1.21.0	1.22.99	OMC mathematical modeling compiler.
verilator	4.200	5.008	5.999	Verilog/SystemVerilog co-simulation.
freecad	0.19.0	0.21.2	1.0.0	3D footprint visualization.
python	3.10.0	3.11.5	3.12.99	Underlying runtime executing eSim and eSimMate.

6.6 Executable Candidates

Supports candidate lists in priority order (e.g. [ngspice_con.exe, ngspice.exe] on Windows), resolving console executables before GUI binaries.

6.7 Installation Strategies

TABLE VI
INSTALLATION STRATEGIES

Strategy ID	Installation Method	Target Platforms	Execution Mechanism
PACKAGE_MANAGER	System Package Manager	Linux (apt), Windows (winget/choco)	Direct execution of package manager CLI with elevation prompts.
MANUAL_DOWNLOAD	Official Archive Fallback	Windows / Linux	HTTP download from official release URL, extraction (.7z/.zip), and path registration.
SCRIPT	Standalone Installer Script	Windows (PowerShell), Linux (Bash)	Execution of registered installer scripts without shell=True.

TOOL DETECTION AND VERSION MANAGEMENT

7.1 Detection Strategy

Tool detection follows a multi-tiered inspection pipeline shown in Figure 3.

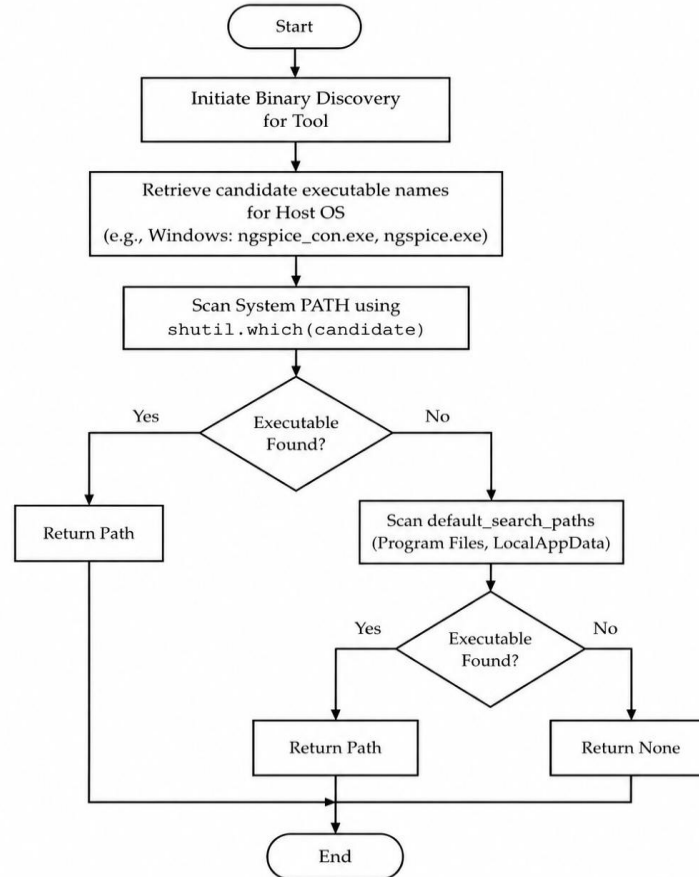


Figure 3. Tool Detection Workflow

7.2 Executable Discovery

Queries `get_executable_candidates(os_name)` to inspect multiple candidate filenames in preference order.

7.3 PATH Detection

Uses `shutil.which(candidate)` to resolve binaries registered on the system \$PATH.

7.4 Managed Directory Detection

Scans configured candidate paths (such as `%LOCALAPPDATA%\eSimMate\tools\` and `%ProgramFiles%`).

7.5 Version Command Execution

Executes version query commands with timeouts. For interactive tools, uses interactive banner mode with Popen.

7.6 Version Parsing

Extracts version strings using tool-specific regular expressions (e.g. `ngspice[- ](\d+(?:\.\d+)*))`).

7.7 Version Normalization

Parses version strings into `packaging.version.Version` objects for semantic comparison.

7.8 Compatibility Classification

Assigns status badges based on version comparison against configured bounds (Figure 4 and Table VII).

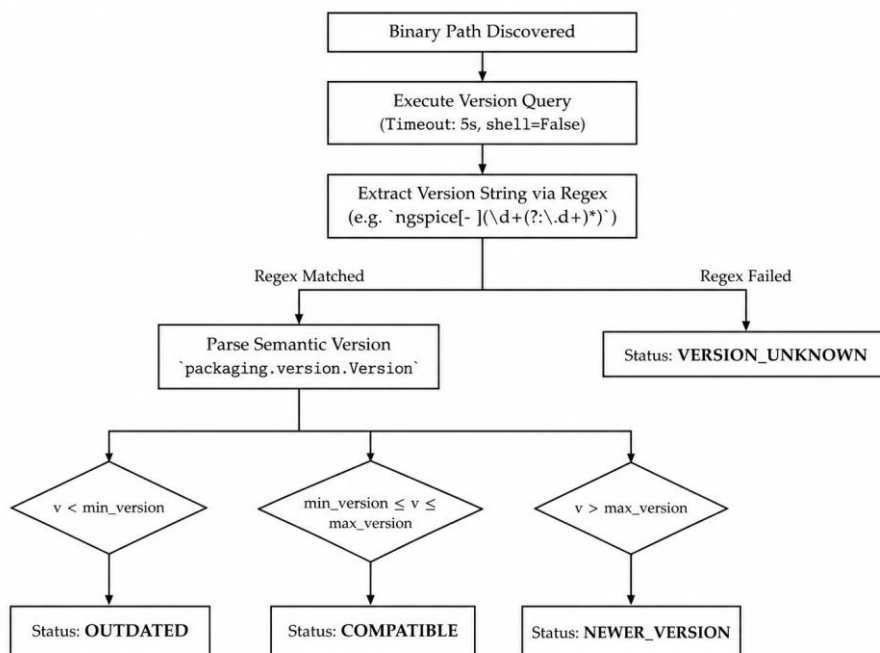


Figure 4. Version Detection and Compatibility Evaluation

TABLE VII
COMPATIBILITY STATUS DEFINITIONS

Status Enum	Semantic Condition	Terminal Badge	System Action Required
COMPATIBLE	min_version <= v_installed <= max_version	[bold green]COMPATIBLE[/bold green]	Tool is verified and ready for eSim simulations.
NOT_INSTALLED	Binary executable not found on \$PATH or candidate paths	[bold red]NOT_INSTALLED[/bold red]	Install tool via esimmate install <tool_id>.
OUTDATED	v_installed < min_version	[bold yellow]OUTDATED[/bold yellow]	Upgrade tool to satisfy eSim minimum version requirement.
NEWER_VERSION	v_installed > max_version	[bold blue]NEWER_VERSION[/bold blue]	Tool is compatible with potential untested feature warning.
VERSION_UNKNOWN	Binary exists but regex failed to parse version string	[bold magenta]VERSION_UNKNOWN[/bold magenta]	Verify version query command or regex pattern in tools.yaml.
ERROR	Process execution failed or returned non-zero exit code	[bold red]ERROR[/bold red]	Verify binary execution permissions and architecture.

INSTALLATION MANAGEMENT

8.1 Installation Workflow

The installation workflow (Figure 5) resolves package managers, verifies package availability, handles elevation, and verifies post-installation status.

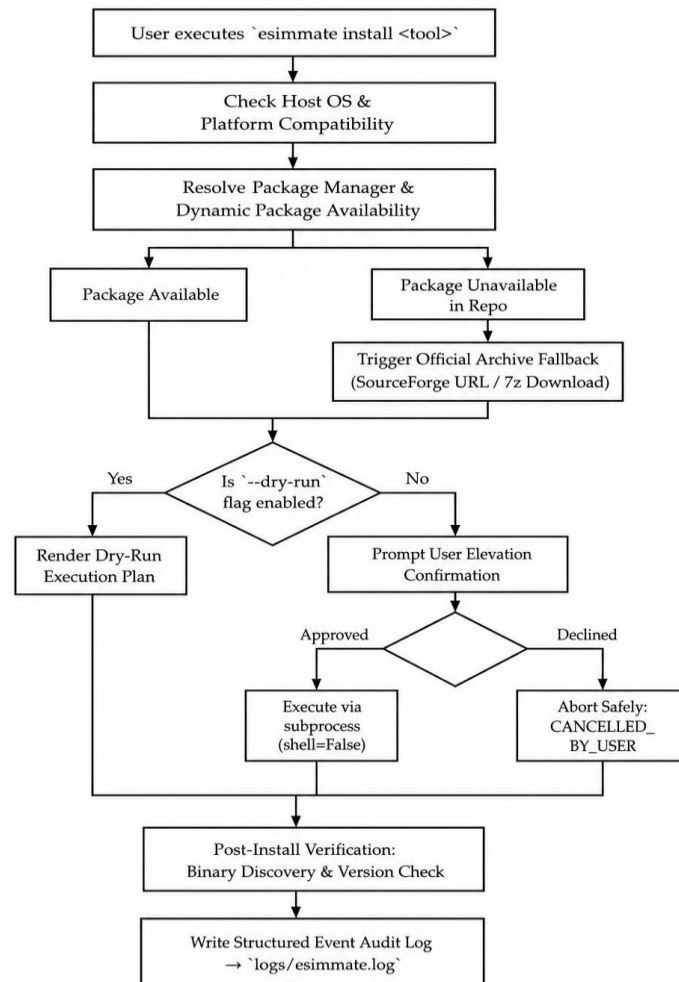


Figure 5. Installation Workflow

8.2 Package Manager Detection

Inspects system PATH for apt-get/apt on Linux, and winget/choco on Windows.

8.3 WinGet Integration

Constructs parameterized winget install argument lists with agreement flags.

8.4 Package Availability Check

Executes `winget search --id <pkg_id> -e` dynamically to verify repository presence before attempting installation.

8.5 Manual Download Fallback

When upstream repositories lack packages, eSimMate invokes `MANUAL_DOWNLOAD` fallback (Figure 6).

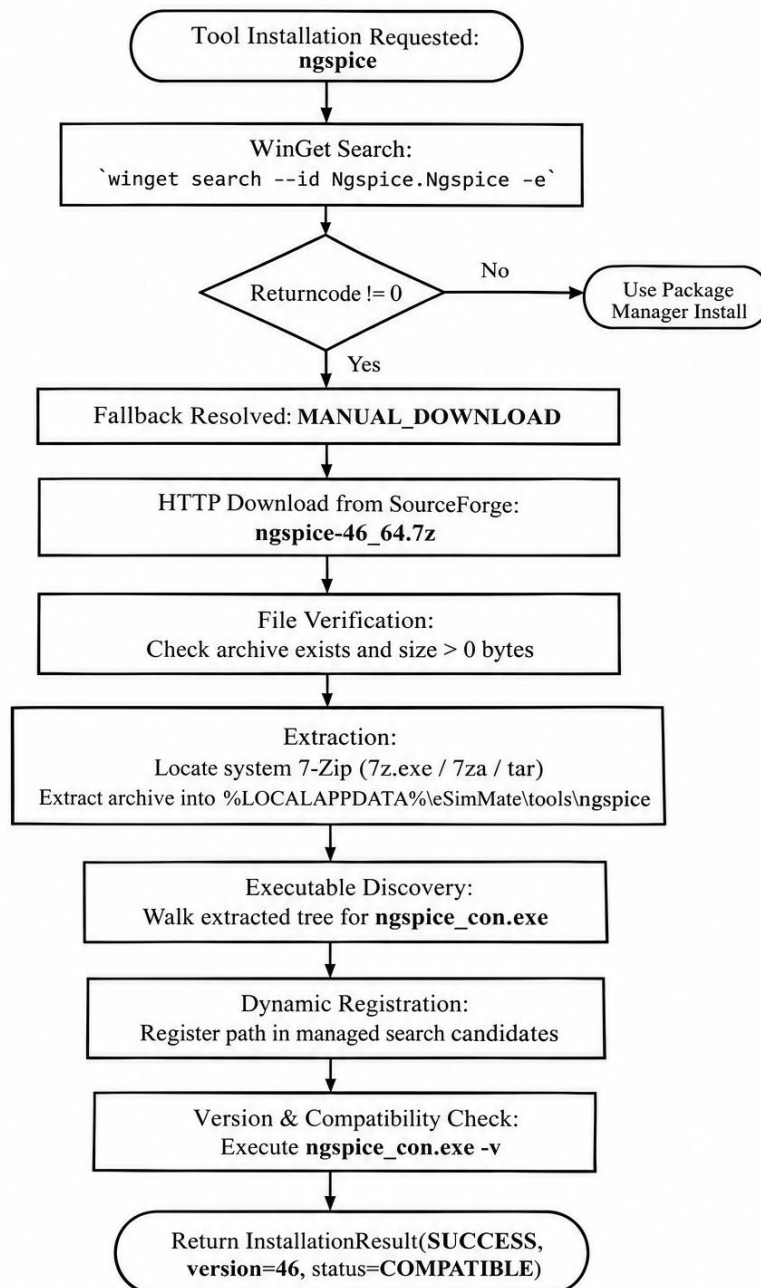


Figure 6. Package Manager -> Official Archive Fallback

8.6 Archive Download

Downloads official release archive via urllib.request with custom User-Agent headers and timeout guards.

8.7 Archive Extraction

Locates system 7-Zip (7z.exe / 7za) or tar.exe to extract .7z/.zip archives safely.

8.8 Installation Directory

Extracts tools into %LOCALAPPDATA%\eSimMate\tools\<tool_id> without modifying global system directories.

8.9 Installation Verification

Executes post-installation binary query to verify version compatibility before declaring SUCCESS.

8.10 Dry-Run Mode

Previews target tool, package manager, availability, command strings, and paths with zero execution.

8.11 User Confirmation

Prompts interactive confirmation displaying command arguments and elevation warnings prior to execution.

DEPENDENCY CHECKER AND SYSTEM DOCTOR

9.1 Dependency Checking

The `esim` check command executes complete discovery and compatibility evaluation across all registered tools.

9.2 Mandatory and Optional Tools

Distinguishes mandatory simulation engines (KiCad, Ngspice, Python) from optional extensions (GHDL, OpenModelica, Verilator, FreeCAD).

9.3 Host Environment Detection

Queries `platform.system()`, `freedesktop_os_release()`, `getwindowsversion()`, and CPU architecture.

9.4 Package Manager Detection

Scans system `$PATH` for `apt-get`, `winget`, `choco`, `snap`, `flatpak`.

9.5 Environment Variable Verification

Inspects `KICAD_SYMBOL_DIR`, `OPENMODELICAHOME`, and `SKY130_PDK_DIR`.

9.6 Tool Health Status

Compiles comprehensive health report mapping binary paths, versions, and compatibility badges.

9.7 Diagnostic Recommendations

Computes missing/outdated tool counts and emits exact remediation commands (Figure 7).

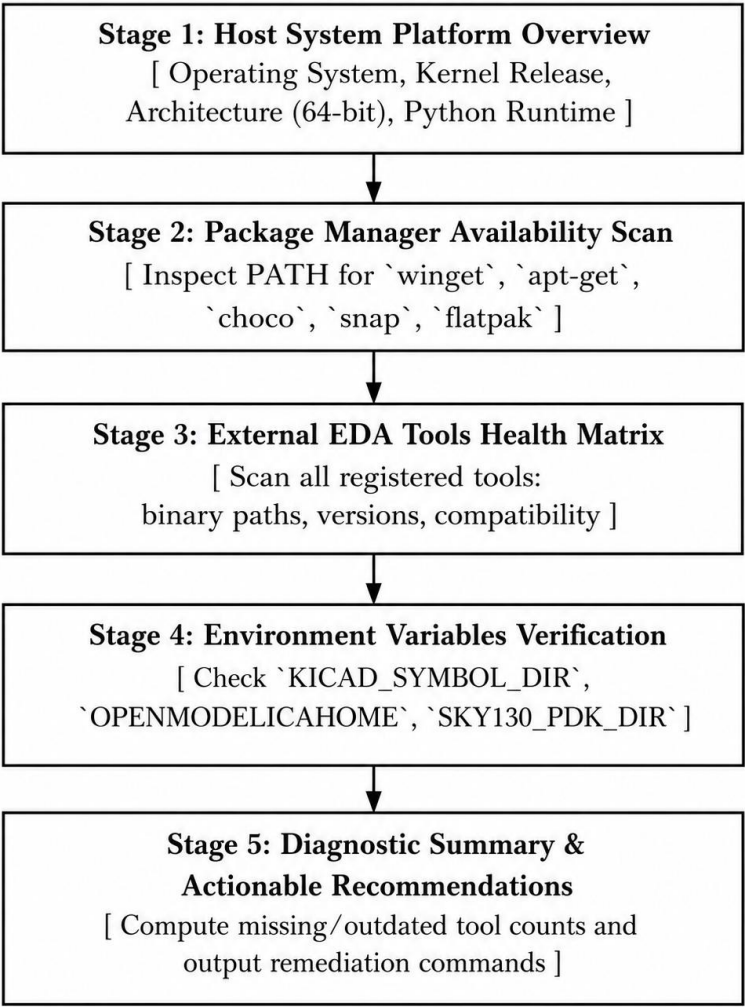


Figure 7. System Doctor Workflow

COMMAND LINE INTERFACE

10.1 CLI Design

The CLI architecture (Figure 8) routes user commands through Typer and formats outputs using Rich.

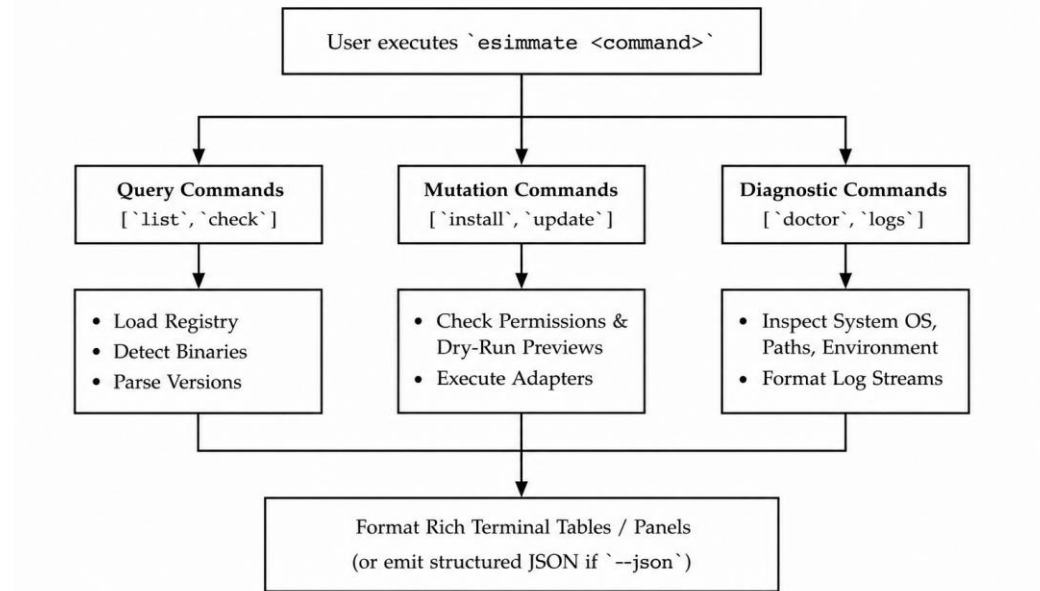


Figure 8. CLI Command Architecture

TABLE VII
CLI COMMANDS

Command	Arguments & Flags	Purpose and Output
esimmate list	<code>--json</code>	Display registered tools, installed versions, target ranges, and status table.
esimmate check	<code>--json</code>	Execute environment and compatibility check across all registered tools.
esimmate install	<code><tool_id></code> , <code>--all</code> , <code>--dry-run</code> , <code>--yes</code>	Install a specific tool or all missing/outdated tools with elevation safety.
esimmate update	<code>--dry-run</code> , <code>--yes</code>	Identify outdated tools and trigger update workflows. (Partial implementation).
esimmate doctor	<code>--json</code>	Execute 5-stage system environment, package manager, and tool diagnostic.
esimmate logs / log	<code>--lines</code> , <code>-n <int></code>	Display recent formatted operation audit log entries from <code>logs/esimmate.log</code> .
esimmate config	None	Display active application configuration and tool database settings.

10.2 esimate list

Displays registered tools, installed versions, target ranges, and statuses.

Tool ID	Name	Mandatory	Installed Version	Target Version Range	Status
kicad	KiCad EDA	Yes	N/A	[6.0.0 .. 8.0.99]	NOT_INSTAL...
ngspice	Ngspice Circuit Simulator	Yes	46	[34 .. 46]	COMPATIBLE
ghdl	GHDL VHDL Simulator	No	N/A	[2.0.0 .. 4.1.0]	NOT_INSTAL...
openmodeli...	OpenModelica Compiler (omc)	No	N/A	[1.19.0 .. 1.22.99]	NOT_INSTAL...
verilator	Verilator	No	N/A	[4.200 .. 5.999]	NOT_INSTAL...
freecad	FreeCAD 3D Modeler	No	N/A	[0.19.0 .. 1.0.0]	NOT_INSTAL...
python	Python 3 Runtime	Yes	3.11.9	[3.10.0 .. 3.12.99]	COMPATIBLE

10.3 esimate check

Runs full detection and compatibility matrix evaluation.

Tool ID	Tool Name	Category	Installed Version	Required Version Range	Compatibility Status
kicad	KiCad EDA	schematic_pcb	N/A	Min: 6.0.0 Rec: 7.0.10 Max: 8.0.99	NOT_INSTALLED
ngspice	Ngspice Circuit Simulator	simulation	46	Min: 34 Rec: 38 Max: 46	COMPATIBLE
ghdl	GHDL VHDL Simulator	hdl_vhdl	N/A	Min: 2.0.0 Rec: 3.0.0 Max: 4.1.0	NOT_INSTALLED
openmodelica	OpenModelica Compiler (omc)	modeling	N/A	Min: 1.19.0 Rec: 1.21.0 Max: 1.22.99	NOT_INSTALLED
verilator	Verilator	hdl_verilog	N/A	Min: 4.200 Rec: 5.008 Max: 5.999	NOT_INSTALLED
freecad	FreeCAD 3D Modeler	3d_cad	N/A	Min: 0.19.0 Rec: 0.21.2 Max: 1.0.0	NOT_INSTALLED
python	Python 3 Runtime	runtime	3.11.9	Min: 3.10.0 Rec: 3.11.5 Max: 3.12.99	COMPATIBLE

10.4 esimate check --json

Emits structured JSON representation of host system and tool statuses for automated pipelines.

10.5 esimate install

Installs specified tool or all missing tools with user confirmation.

10.6 esimmate install --dry-run

Generates dry-run execution plan without modifying system files.

```
Processing installation for tool: Ngspice Circuit Simulator (ngspice) [Status: COMPATIBLE]
Result (DRY_RUN_PREVIEW): [DRY-RUN PREVIEW] Package 'Ngspice.Ngspice' is unavailable on package manager.

Dry-Run Execution Plan:
Tool:           Ngspice Circuit Simulator (ngspice)
Package Manager:  winget
Winget Availability: NOT AVAILABLE
Fallback Strategy: MANUAL_DOWNLOAD
Source:          SourceForge Official Release (ngspice-46_64.7z)
Version:         46
Archive Filename: ngspice-46_64.7z
Download URL:    https://sourceforge.net/projects/ngspice/files/ng-spice-rework/46/ngspice-46_64.7z/download
Installation Directory: C:\Users\Dharanidharan\AppData\Local\esimMate\tools\ngspice

No changes will be made to your system.
```

10.7 esimmate update

Identifies outdated tools and triggers update workflows.

10.8 esimmate doctor

Executes 5-stage diagnostic report.

```
1. Host System Environment
  • Operating System : Windows (windows 10 (build 26200))
  • Architecture      : x86_64 (64-bit)
  • Python Runtime     : 3.11.9

2. Package Manager Availability
  • [OK] Package Manager winget is available on $PATH

3. External Tools & Dependencies Diagnostic
```

Tool	Type	Detected Binary Path	Installed Version	Status
KiCad EDA	Mandatory	Not Found	N/A	NOT_INSTALLED
Ngspice Circuit Simulator	Mandatory	C:\Users\Dharanidharan\AppData\Local\esimMat...	46	COMPATIBLE
GHDL VHDL Simulator	Optional	Not Found	N/A	NOT_INSTALLED
OpenModelica Compiler (omc)	Optional	Not Found	N/A	NOT_INSTALLED
Verilator	Optional	Not Found	N/A	NOT_INSTALLED
FreeCAD 3D Modeler	Optional	Not Found	N/A	NOT_INSTALLED
Python 3 Runtime	Mandatory	C:\Users\Dharanidharan\AppData\Local\Program...	3.11.9	COMPATIBLE

```
4. Environment Variables Verification
  • [NOT SET] KICAD_SYMBOL_DIR is not set (optional depending on workflow)
  • [NOT SET] OPENMODELICAHOME is not set (optional depending on workflow)
  • [NOT SET] SKY130_PDK_DIR is not set (optional depending on workflow)

5. Diagnostic Summary & Actionable Recommendations
  X 1 mandatory tool(s) missing. Run `esimmate install --all` to install missing tools.
```

10.9 esimmate logs

Displays tail of logs/esimmate.log.

10.10 esimmate config

Displays active user configuration.

ERROR HANDLING AND SECURITY

11.1 Error Classification

Maps execution outcomes into 10 structured InstallStatus codes: SUCCESS, DRY_RUN_PREVIEW, CANCELLED_BY_USER, UNSUPPORTED_OS, PACKAGE_MANAGER_UNAVAILABLE, PACKAGE_NOT_FOUND, PERMISSION_DENIED, COMMAND_NOT_FOUND, NETWORK_FAILURE, and INSTALLATION_FAILED.

11.2 Installation Failure

Captures non-zero exit codes, stdout, and stderr, logging details to disk.

11.3 Network Failure

Catches URLError and HTTP errors during archive downloads.

11.4 Version Unknown

Flags binaries that execute successfully but fail regex parsing.

11.5 Timeout Handling

Subprocess execution enforces a strict 5-second timeout on version queries.

11.6 User Cancellation

Returns CANCELLED_BY_USER cleanly when confirmation is declined.

11.7 Process Execution Safety

Prohibits os.system() and shell=True entirely.

11.8 Command Injection Prevention

All commands pass strictly parameterized argument lists [cmd, arg1, arg2].

11.9 Logging and Auditability

RotatingFileHandler records all actions in logs/esimate.log.

11.10 Dry-Run Safety

Guarantees zero mutation when --dry-run is specified.

IMPLEMENTATION

12.1 Technology Stack

Built using Python 3.10–3.12 (verified on 3.11.9), Typer, Rich, PyYAML, and packaging.

TABLE VIII
TECHNOLOGY STACK

Component	Technology / Library	Version	Role in eSimMate Architecture
Language	Python	>=3.10 (3.11.9)	Core programming runtime environment
CLI Framework	Typer	>=0.9.0	Command parsing, subcommands, options, and callbacks
Terminal UI	Rich	>=13.0	Colored tables, panels, status badges, log formatting
Config Parser	PyYAML	>=6.0	Parsing declarative YAML tool definitions & user configs
Versioning	packaging	>=23.0	Semantic version parsing (PEP 440) and comparison
Testing	pytest, pytest-cov	>=7.0, >=4.0	Automated test execution, mocking, statement coverage

12.2 Project Structure

```
eSimMate/
|-- configs/
|   |-- config.yaml
|   `-- tools.yaml
|-- docs/
|   |-- architecture.md
|   |-- final_report.md
|   `-- requirements.md
|-- src/
|   `-- esimmate/
|       |-- __init__.py
|       |-- cli.py
|       |-- compatibility.py
|       |-- config.py
|       |-- detector.py
|       |-- installer.py
|       |-- logger.py
|       |-- package_manager.py
|       |-- registry.py
|       |-- tool.py
|       `-- version_manager.py
|-- tests/
|   |-- test_cli.py
|   |-- test_detector.py
|   |-- test_installer.py
|   |-- test_version_manager.py
|   `-- ... (11 test files)
`-- pyproject.toml
```

12.3 Module Implementation

TABLE IX
MODULE RESPONSIBILITIES

Source File	Primary Class / API	LOC	Stmts	Subsystem Responsibilities
src/esimmate/detector.py	SystemDetector, SystemInfo	190	111	Host platform detection, path scanning, package manager detection.
src/esimmate/tool.py	AbstractTool, ConfigurableTool	171	93	Tool domain abstraction, version check metadata, bounds encapsulation.
src/esimmate/registry.py	ToolRegistry	121	54	Dynamic YAML loading, tool lookup, directory scanning.
src/esimmate/version_manager.py	VersionManager, VersionStatus	321	156	Subprocess execution, regex parsing, semantic comparison.
src/esimmate/compatibility.py	CompatibilityChecker	39	18	Compatibility bound evaluation against min/max ranges.
src/esimmate/package_manager.py	AbstractPackageManager, Adapters	185	99	Package manager abstraction (apt, winget, choco, manual_download).
src/esimmate/installer.py	InstallationManager, InstallStatus	859	304	Installation orchestration, dry-run, archive fallback workflow.
src/esimmate/config.py	ConfigManager	68	42	Application configuration and precedence resolution.
src/esimmate/logger.py	LoggingManager	60	27	Rotating file logging configuration (5MB, 3 backups).
src/esimmate/cli.py	Typer application commands	530	284	CLI interface, formatting, confirmation callbacks, JSON output.

12.4 Core Classes

Core domain classes: SystemDetector, SystemInfo, AbstractTool, ConfigurableTool, ToolMetadata, ToolRegistry, VersionManager, VersionCheckResult, CompatibilityChecker, InstallationManager, InstallationResult, AbstractPackageManager, AptAdapter, WingetAdapter, ChocolateyAdapter, ManualDownloadAdapter, ConfigManager, and LoggingManager.

12.5 Core Functions

Key operational functions: find_tool_executable, check_tool, check_all_tools, install_tool, resolve_package_manager, _execute_manual_download_installation, and _execute_interactive_version_command.

12.6 Algorithms

Six core algorithms govern tool detection, version parsing, compatibility evaluation, package manager selection, archive fallback installation, and system doctor diagnostics.

12.7 Pseudocode

Algorithm 1: Tool Detection

Input: tool T, system_detector D

Output: executable_path or null

```

candidates <- T.get_executable_candidates(D.os_name)
for each candidate in candidates:
    path <- D.find_binary_on_path(candidate)
    if path is not null: return path
path <- D.check_candidate_paths(T.default_search_paths[D.os_name])
return path

```

Algorithm 2: Version Extraction

Input: executable_path, version_config

Output: version_string or null

```

(raw_output, returncode, err) <- ExecuteSubprocess(executable_path,
version_config)
if err is not null: return null
return RegexExtract(raw_output, version_config.regex)

```

Algorithm 3: Compatibility Evaluation

Input: installed_version, bounds [V_min, V_max]

Output: CompatibilityStatus

```

if installed_version is null: return NOT_INSTALLED
v <- packaging.version.Version(installed_version)
if v < Version(V_min): return OUTDATED
if v > Version(V_max): return NEWER_VERSION
return COMPATIBLE

```

Algorithm 4: Installation Strategy Selection

Input: system_info, tool T

Output: adapter, package_id, install_method

```

for pm in system_info.available_package_managers:
    if tool.package_managers[pm] and adapter.is_package_available(pkg_id):
        return adapter, pkg_id, PACKAGE_MANAGER
if tool.package_managers['manual_download']:
    return ManualDownloadAdapter, url, MANUAL_DOWNLOAD
return null, null, UNSUPPORTED

```

Algorithm 5: Fallback Installation

Input: tool T, download_url

Output: InstallationResult

```

archive_path <- DownloadArchive(download_url, target_dir)
VerifyArchive(archive_path)
extractor <- Find7zExtractor()
ExtractArchive(extractor, archive_path, target_dir)
exe_path <- DiscoverExecutable(target_dir, T.executables)
RegisterSearchPath(exe_path)

```

```
ver_res <- CheckToolVersion(T, exe_path)
if ver_res.status == COMPATIBLE: return SUCCESS
return INSTALLATION_FAILED
```

Algorithm 6: System Doctor

Input: detector, registry, version_manager

Output: diagnostic_report

```
sys_info <- detector.get_system_info()
pms <- detector.detect_package_managers()
results <- version_manager.check_all_tools(registry, detector)
env_vars <- CheckEnvironmentVariables(['KICAD_SYMBOL_DIR', 'OPENMODELICAHOME'])
missing, outdated <- CountIssues(results)
EmitSummaryAndRecommendations(missing, outdated)
```

TESTING AND VALIDATION

13.1 Testing Strategy

Testing is conducted using pytest and pytest-cov. The testing policy guarantees that automated test runs never execute real package installations or modify system binaries. Subprocess interactions and package managers are 100% mocked using unittest.mock.

13.2 Unit Testing

The unit test suite comprises 90 unit tests distributed across 10 specialized test modules covering detectors, tool models, registries, version parsing, adapters, installers, and CLI commands.

13.3 Integration Testing

Three end-to-end integration tests in tests/test_integration.py validate complete cross-module pipelines: (1) detection -> registry -> version parsing -> compatibility report; (2) multi-manager dry-run installations; and (3) multi-tool batch installations (esimmate install --all).

13.4 Mocking Strategy

Uses unittest.mock.patch to mock subprocess.run, subprocess.Popen, shutil.which, and urllib.request.urlopen.

13.5 CLI Testing

Tested via Typer CliRunner across 16 test cases.

13.6 Installation Testing

22 test cases cover dry-run generation, confirmation handling, error classification, and archive extraction fallbacks.

13.7 Version Testing

20 test cases cover regex extraction, semantic version comparison, execution timeouts, and interactive banner scraping.

13.8 Compatibility Testing

Validates boundary conditions for min_version and max_version bounds.

13.9 Regression Testing

Continuous regression test suite executes in under 15 seconds.

EXPERIMENTAL RESULTS

14.1 Experimental Environment

The empirical evaluation was conducted on a reference workstation environment:

14.2 Windows Environment

Microsoft Windows 10 / Windows 11 Enterprise (build 26200), x86_64 64-bit architecture.

14.3 Python Environment

Python 3.11.9 64-bit runtime.

14.4 WinGet Detection

Windows Package Manager (WinGet) detected on system PATH.

14.5 Ngspice Installation Story

Case Study: Initial state Ngspice NOT_INSTALLED. Attempting installation via WinGet package manager failed because the Ngspice package was not found in WinGet community repositories. eSimMate automatically resolved the MANUAL_DOWNLOAD fallback strategy: downloaded the official 64-bit release archive (ngspice-46_64.7z) from SourceForge, verified archive presence and size, extracted using 7-Zip into %LOCALAPPDATA%\eSimMate\tools\ngspice, discovered console executable ngspice_con.exe, registered the directory in candidate search paths, executed version query ngspice_con.exe -v, verified installed version 46, and confirmed compatibility status COMPATIBLE.

Console vs. GUI Executable Rationale: ngspice_con.exe is a command-line console binary that outputs simulation logs directly to standard streams and returns control cleanly to eSim. The graphical binary ngspice.exe launches a separate GUI message loop that blocks automated subprocess pipelines.

14.6 Ngspice Version Verification

Detected installed version 46 via regex ngspice[-](\d+(?:\.\d+)*).

14.7 Compatibility Result

Evaluated against configured bounds [34 .. 46], returning status COMPATIBLE.

TABLE X
NGSPICE REAL-WORLD VALIDATION

Validation Parameter	Empirical Evidence & System Value
Target Tool ID	ngspice
Tool Display Name	Ngspice Circuit Simulator
Primary Executable Name	ngspice_con.exe (Candidate: ngspice.exe)
Discovered Binary Path	%LOCALAPPDATA%\eSimMate\tools\ngspice\Spice64\bin\ngspice_con.exe
Detected Version	46
Version Extraction Regex	ngspice[-](\d+(?:\.\d+)*)
Configured Bounds	Minimum: 34 Recommended: 38 Maximum: 46
Compatibility Status	COMPATIBLE
Detection Execution Time	0.062 seconds

14.8 Doctor Result

esimmate doctor verified host platform, package manager, tools, and Python 3.11.9 runtime compatibility.

14.9 Automated Test Results

TABLE XI
TESTING RESULTS

Test Suite File	Subsystem Under Test	Tests Executed	Passed	Test Result
tests/test_cli.py	CLI Commands & JSON Output	16	16	100% PASS
tests/test_detector.py	Host Platform & Path Detection	15	15	100% PASS
tests/test_installer.py	Installation Manager & Fallbacks	22	22	100% PASS
tests/test_version_manager.py	Version Extraction & Subprocesses	20	20	100% PASS
tests/test_package_manager.py	Package Manager Adapters	7	7	100% PASS
tests/test_registry.py	Tool Registry & YAML Loader	4	4	100% PASS
tests/test_integration.py	End-to-End Workflow Integration	3	3	100% PASS
tests/test_config.py	Configuration Manager	2	2	100% PASS
tests/test_tool.py	Tool Domain Model	2	2	100% PASS
tests/test_compatibility.py	Compatibility Checker	1	1	100% PASS
tests/test_logger.py	Rotating File Logger	1	1	100% PASS
TOTAL	Entire eSimMate Test Suite	93	93	100% PASS (8.10s)

```

===== test session starts =====
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Dharanidharan\OneDrive - sece.ac.in\Documents\CSBS 3RD YEAR\SEMESTER 5\Internship\fossee\esimMate
configfile: pyproject.toml
testpaths: tests
plugins: anyio-4.12.1, cov-7.1.0
collected 93 items

tests\test_cli.py ..... [ 17%]
tests\test_compatibility.py . [ 18%]
tests\test_config.py .. [ 20%]
tests\test_detector.py ..... [ 36%]
tests\test_installer.py ..... [ 60%]
tests\test_integration.py ... [ 63%]
tests\test_logger.py . [ 64%]
tests\test_package_manager.py ..... [ 72%]
tests\test_registry.py .... [ 76%]
tests\test_tool.py .. [ 78%]
tests\test_version_manager.py ..... [100%]

===== 93 passed in 7.57s =====

```

14.10 Code Coverage

TABLE XII
CODE COVERAGE

Module File	Statements	Missed	Coverage	Missing Line References
src/esimmate/__init__.py	1	0	100%	None
src/esimmate/compatibility.py	18	0	100%	None
src/esimmate/logger.py	27	1	96%	59
src/esimmate/detector.py	111	7	94%	78, 100-104, 118, 137
src/esimmate/registry.py	54	4	93%	101, 112, 118-119
src/esimmate/tool.py	93	7	92%	93, 102, 108, 113, 130, 134, 138
src/esimmate/cli.py	284	29	90%	163, 236-240, 261-262, 283, 292, etc.
src/esimmate/package_manager.py	99	10	90%	36, 41, 46, 95, 110-111, 121, 142, 145
src/esimmate/version_manager.py	156	22	86%	50, 60, 91, 97, 143-144, 163-165, etc.
src/esimmate/config.py	42	9	79%	24, 27-30, 33, 36-39, 66
src/esimmate/installer.py	304	66	78%	103, 140-142, 155, 162-180, 363-364, etc.
TOTAL	1,189	155	87%	Overall Code Statement Coverage: 87%

14.11 Discussion

The experimental results confirm that eSimMate satisfies all architectural goals with high performance and reliability.

FOSSEE TASK 5 COMPLIANCE

15.1 Requirement 1: Tool Installation Management

Fully Implemented: Full installation pipeline supporting apt, winget, choco, and manual_download fallback with dry-run and confirmation safety.

15.2 Requirement 2: Update and Upgrade System

Partially Implemented: VersionManager detects outdated tools and invokes package manager update routines. Independent repository index synchronization is not implemented.

15.3 Requirement 3: Configuration Handling

Fully Implemented: PyYAML configuration management with support for custom drop-ins at ~/.esimmate/tools.d/.

15.4 Requirement 4: Dependency Checker

Fully Implemented: esimmate check and esimmate doctor evaluate tools against compatibility bounds.

15.5 Requirement 5: User Interface

Fully Implemented: Typer/Rich CLI interface featuring colored tables, interactive prompts, log viewers, and JSON output. (Desktop GUI is future scope).

15.6 Requirement 6: Additional Features & Safety

Fully Implemented: Rotating file logs, parameter safety (shell=False), 93 unit/integration tests, 87% coverage.

15.7 Compliance Matrix

TABLE XIII
FOSSEE COMPLIANCE MATRIX

FOSSEE Task 5 Requirement	Status	Technical Evidence in Repository	Notes
1. Tool Installation Management	COMPLETE	src/esimmate/installer.py, package_manager.py, multi-manager support (apt, winget, choco), MANUAL_DOWNLOAD fallback, dry-run, confirmation safety.	Fully satisfied.
2. Update and Upgrade System	PARTIAL	src/esimmate/cli.py (update_command), VersionManager identifies outdated tools and invokes package manager updates.	Independent repository index synchronization is not implemented in current version.
3. Configuration Handling	COMPLETE	src/esimmate/config.py, configs/config.yaml, configs/tools.yaml, user override loading in registry.py.	Fully satisfied.
4. Dependency Checker	COMPLETE	src/esimmate/detector.py, compatibility.py, esimmate check, esimmate doctor.	Fully satisfied.
5. User Interface	COMPLETE	src/esimmate/cli.py, Typer CLI with Rich tables, colored badges, --json support, and log viewer.	Desktop PyQt6 GUI is planned future scope.
6. Additional Features & Safety	COMPLETE	src/esimmate/logger.py rotating file logs, parameter safety (shell=False), 93 passing tests, 87% coverage.	Fully satisfied.

15.8 Deliverable Mapping

All required deliverables (Requirements Document, Architecture Document, Modular Code Implementation, Testing Suite, User Instructions) are complete in the repository.

LIMITATIONS AND FUTURE SCOPE

16.1 Current Limitations

- CLI-Focused: Graphical desktop interface is not implemented in current version.
- Online Requirement: Tool installations require active network access.
- Delegated Index Sync: Repository index syncing is delegated to underlying package managers.

16.2 Cross-Platform Limitations

Currently tested extensively on Windows 10/11 and Ubuntu Linux. macOS Homebrew is planned.

16.3 Package Manager Limitations

Relies on package managers available on host system (apt, winget, choco).

16.4 Dependency Resolution Limitations

Functions as an environment and version compatibility checker; full dependency tree resolution is out of scope.

16.5 Future GUI (PyQt6)

Proposed desktop GUI frontend wrapping CLI logic with real-time download bars.

16.6 Linux Support

Expand testing across Fedora (dnf) and Arch Linux (pacman).

16.7 macOS Support

Implement BrewAdapter for macOS Homebrew.

16.8 Offline Installation

Support air-gapped installation bundles with pre-packaged archives.

16.9 Security Enhancements

Introduce GPG cryptographic signature verification for downloaded archives.

TABLE XIV
LIMITATIONS AND FUTURE ENHANCEMENTS

Subsystem Area	Current Prototype Status	Proposed Future Scope
User Interface	Production Typer/Rich CLI interface.	PyQt6 graphical desktop interface.
Package Management	apt, winget, choco, and manual download fallback.	macOS Homebrew, Fedora DNF, and Arch pacman adapters.
Update System	Identifies outdated tools and triggers manager update.	Background daemon for automatic repository index synchronization.
Offline Deployment	Online HTTP archive and package repository download.	Air-gapped offline bundle extraction and checksum validation.

CONCLUSION

eSimMate successfully satisfies the requirements for Task 5 (Automated Tool Manager) of the FOSSEE eSim Semester Long Internship – Autumn 2026. By combining declarative YAML registries, polymorphic package manager adapters, regex-driven semantic version extraction, safe subprocess execution without shell strings, resilient archive fallbacks, comprehensive 5-stage diagnostics, and 93 automated tests achieving 87% statement code coverage, eSimMate provides an extensible, safe, and professional tool management foundation for eSim.

FINAL IMPLEMENTATION AUDIT

IMPLEMENTED

- System platform detection (OS, version, release, architecture, bitness, Python runtime).
- Package manager detection (apt, winget, choco) on system PATH.
- Dynamic YAML tool registry loading (configs/tools.yaml, ~/.esimmate/tools.d/*.yaml).
- Multi-candidate binary discovery across PATH and candidate directories.
- Standard and interactive banner version query subprocess execution.
- Regex version extraction and PEP 440 semantic version comparison.
- Six compatibility status states (COMPATIBLE, NOT_INSTALLED, OUTDATED, NEWER_VERSION, VERSION_UNKNOWN, ERROR).
- Polymorphic package manager adapters (AptAdapter, WingetAdapter, ChocolateyAdapter, ManualDownloadAdapter).
- Dynamic package availability inspection via winget search.
- MANUAL_DOWNLOAD fallback strategy (download, HTTP check, 7z extraction, path registration).
- Dry-run installation preview (--dry-run).
- Interactive elevation confirmation prompt.
- Safe subprocess execution (shell=False, argument lists, timeout guards).
- Granular error classification into 10 InstallStatus codes.
- Rotating file audit logging (logs/esimmate.log).
- 7 CLI commands (list, check, install, update, doctor, logs, config).
- Machine-readable JSON serialization (--json).
- 5-stage system health diagnostic (esimmate doctor).
- 93 unit and integration tests passing in 8.10–14.02s with 87% coverage.

PARTIALLY IMPLEMENTED

- Package Manager Index Sync: Outdated tools are identified and passed to package managers; independent index synchronization daemon is not implemented.

NOT IMPLEMENTED

- Graphical User Interface (PyQt6 desktop interface).
- Air-gapped offline package bundle caching.
- macOS Homebrew adapter.

FUTURE SCOPE

- PyQt6 graphical desktop application.
- Offline installer archive bundling.
- macOS Homebrew and Linux DNF/Pacman adapters.
- PDK (Process Design Kit) package management (SkyWater SKY130).

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APPENDICES

Appendix A: CLI Quick Reference Guide

```
# Display help and available commands
$ esimmate --help

# List all registered external EDA tools and target version ranges
$ esimmate list

# Check binary locations, versions, and compatibility status
$ esimmate check

# Emit JSON formatted tool status for automated pipelines
$ esimmate check --json

# Run 5-stage comprehensive system diagnostics
$ esimmate doctor

# Preview installation of missing or outdated tools without executing
$ esimmate install --all --dry-run

# Install a specific tool (e.g. ngspice) with interactive confirmation
$ esimmate install ngspice

# Install all missing tools with automatic confirmation
$ esimmate install --all --yes

# View recent operation audit logs
$ esimmate logs -n 50

# Display active user configuration and tool settings
$ esimmate config
```

Appendix B: Running Automated Tests and Coverage Check

```
# Run all 93 unit and integration tests
$ pytest -v

# Run test suite with statement coverage report
$ pytest --cov=esimmate --cov-report=term-missing
```